Systems Verification
Exercise Sheet 05

Exercise 1: Understanding Linear Temporal Logic

Are the following equivalences of LTL valid? If yes, prove both directions of the implication, if not, give a counter-example.

1. $\Diamond p \wedge q \longleftrightarrow \Diamond(p \wedge q)$,
2. $\Box p \wedge \Box q \longleftrightarrow \Box(p \wedge q)$,
3. $\Diamond(q \mathcal{U} p) \longleftrightarrow \Diamond q$,

Exercise 2: Never claim and the corresponding LTL formula

1. Let q be an atomic proposition for some model and consider the LTL formula $\Diamond\Box q$. Write your own corresponding *never claim* that assures satisfiability of the formula in the model.
2. Use the **-f** spin command and compare this to the solution you gave to the same LTL formula in the preivous step.

Exercise 3: Formalizing properties with LTL formulas

Let q, p be an atomic proposition for some model. Construct LTL formulas that encode the following properties of our system (where "time" is measured as execution steps):

- p must be true at the 3^{rd} execution step.
- p must be true at ONLY the 3^{rd} execution step.
- q needs to hold for every execution step and p has to be true infinitely often during the execution of our system.
- The first time in our execution that q is true, we know that from the next execution step onwards, p will never be false.
- At any execution point, if q is true, p has to be false and vice-versa.

Deadline: Tuesday 6^{th} May 2025, 8:00 a.m.